

Dor Alagem

Software Engineering Student

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PROFILE

Computer Science student (GPA 95) and incoming M.Sc. student in Machine Learning & Big Data, with hands-on experience building Python-based backend systems, distributed applications, and automation workflows. Strong foundations in software engineering, operating systems, computer networks, and algorithmic problem solving.

SKILLS

Programming Languages

Python, C, C++, SQL

Backend & Systems

REST APIs, Distributed Systems, MQTT, Automation Workflows, System Validation

Data Analysis

NumPy, Pandas, Scikit-learn

Cloud & Tools

AWS, Git, GitHub, VS Code, PyCharm

EDUCATION

M.Sc. in Computer Science – Machine Learning & Big Data

The Open University of Israel | Incoming (Oct 2026)

B.Sc. in Computer Science

HIT - Holon Institute of Technology

04/2024 – 10/2026

- **GPA: 95**
- **Distinguished Coursework:** Machine Learning (100); Object-Oriented Programming (100); Operating Systems (90); Computer Networks (94)
- **Dean's List (2024–2025)** - Awarded for academic excellence and consistent high performance.

PROJECTS

Smart IoT Irrigation System

Technologies: **Python, MQTT, JSON**

- Designed and implemented a **Distributed IoT System** for **Real-Time Monitoring** and sensor coordination.
- Built an **MQTT-based Communication Architecture** for telemetry exchange between sensor nodes and a central service.
- Developed a dashboard for real-time visualization of environmental sensor data.
- Designed modular components to improve **Scalability** and maintainability.

Autonomous Path Planning Visualizer

Technologies: **Python, NumPy, Tkinter, A* Algorithm**

- Implemented the **A* Pathfinding Algorithm** to compute optimal collision-free routes.
- Developed **Obstacle Avoidance Logic** under dynamic constraints and safety margins.
- Built an interactive visualization tool to demonstrate route selection and algorithm behavior.
- Optimized path calculations using **NumPy** for efficient matrix operations.